# The Atom

## 1. What is an atom?

Atoms are the building blocks that make up all the matter around us: the ground, the air and all gases, water and all liquids, food, chemical elements, and even our own body. Everything around us, including ourselves, is made of atoms.

**!** Very important

The atom is the smallest unit of matter that retains the properties of an element. All the atoms known to humanity are organized in the Periodic Table.

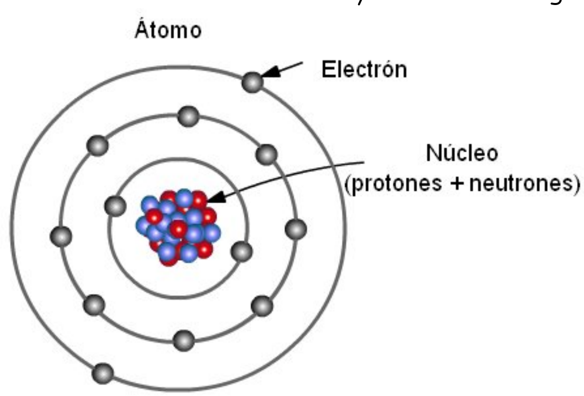
Atoms are extremely small (nanometric scale, between $0,1 \cdot 10^{-9} m$ and $0,5 \cdot 10^{-9} m$): they cannot be seen with the naked eye or with an optical microscope.

**i** Learn more

Atoms can only be seen with special microscopes such as: Scanning Tunneling Microscope (STM), Atomic Force Microscope (AFM), High-resolution Transmission Electron Microscope (TEM/STEM).

## 2. Subatomic particles

Here we show a model of the atom based on the model proposed by Bohr. We say it is a model because the atom is actually much stranger. According to this model, the atom can be drawn like this:




Atoms are made up of three main particles:

| Particle | Symbol in notes | Electric charge | Mass (approx.) | Location |
|---|---|---|---|---|
| Proton | $p^+$ | +1 | 1 | Nucleus |
| Neutron | $n$ | 0 | 1 | Nucleus |
| Electron | $e^-$ | -1 | ~0 | Orbit around the nucleus in the electron cloud. |

- Protons and neutrons are in the nucleus.
- Electrons move around the nucleus in orbitals like planets orbiting the Sun.
- Neutrons and protons are much heavier than electrons. Therefore, when calculating mass, we only add protons and neutrons.

# 3. Very important numbers

### Atomic number (Z)

The atomic number (Z) is the number of protons in an atom.
- It defines the element.
- Example: Hydrogen has Z = 1. This can be seen in the periodic table.

$$Z = n^{\circ}\ p^{+}$$

Where $n^{\circ}\ p^{+}$ = number of protons.

### Mass number (A)

The mass number (A) is the sum of the number of protons and neutrons. It tells us the mass of a specific atom (remember that electrons are so light that we ignore them for mass).

$$A = Z + n\ n$$

Where:
- Z = number of protons.
- $n\ n$ = number of neutrons.

### Electric charge (Q)

To calculate the electric charge, you only need to subtract the number of electrons from the number of protons.

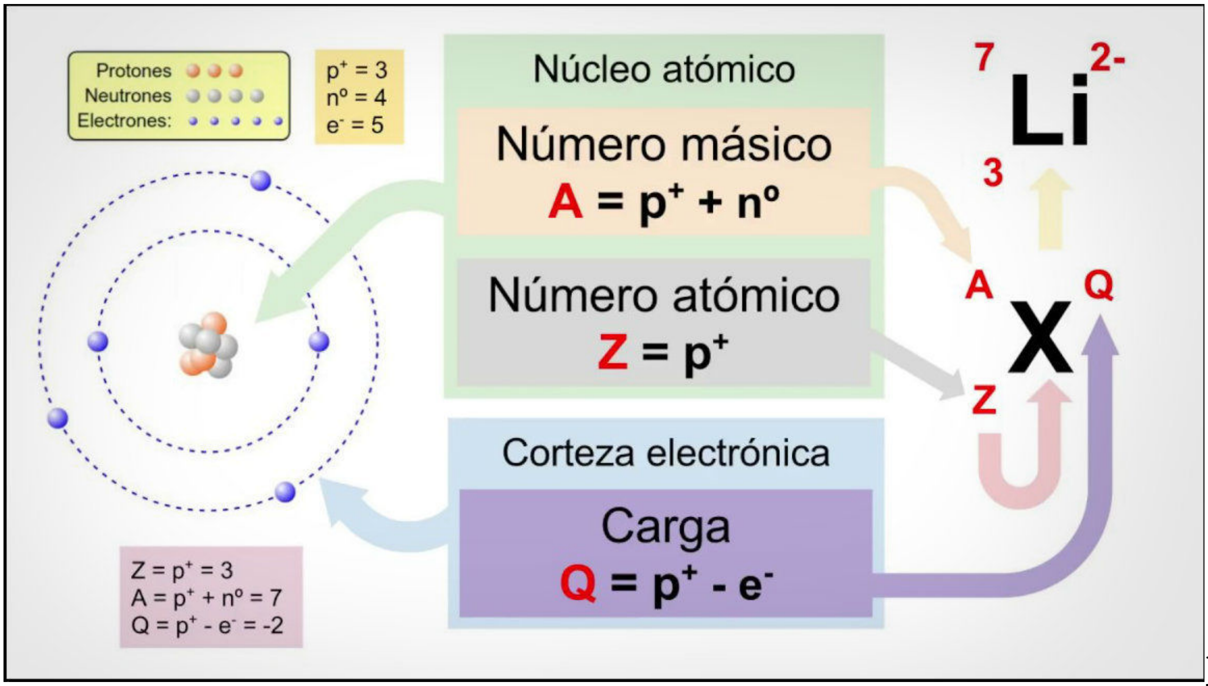
$$Q = n\ p^{+}\ +\ n\ e^{-}$$

If Q is 0, we say the atom is neutral, which means it has no charge.
If Q is positive, the atom has a positive charge.
If Q is negative, the atom has a negative charge.

> **Learn more**
>
> Video explaining electric charge

> **Very important**
>
> Summary of important numbers:
> 1. Atomic number: $Z = n\ p^{+}$
> 2. Mass number: $A = Z + n\ n$
> 3. Electric charge: $Q = n\ p^{+} + n\ e^{-}$

With these numbers, we usually identify an atom in a very simplified way like this:




] # 4. Isotopes

Isotopes are atoms of the same element with:

- The same number of protons (Z)
- A different number of neutrons ($n\ n$)

An example is carbon isotopes: Carbon-12 and Carbon-14. They are written as: *element-A*.

Example of hydrogen isotopes:

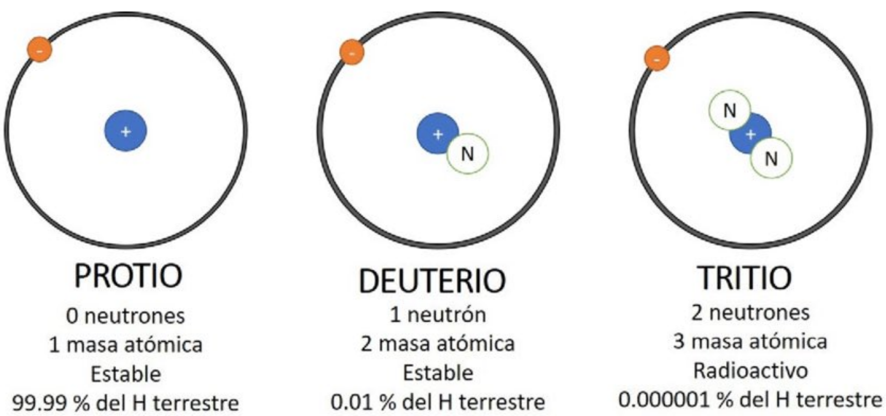



Figure 1: Hydrogen isotopes

# 5. Ions

Ions are atoms that have an electric charge, meaning the number of protons is different from the number of electrons. Let's break it down:
- A neutral atom is the usual type of atom. This means its electric charge is zero. This happens when we have the same number of protons ($p^+$) and electrons ($e^-$). Each positive charge of a proton is balanced by a negative charge of an electron, making the total charge zero.
- An atom that does have an electric charge (that is: $Q \neq 0$) is called an ion. Remember this happens when: protons $\neq$ electrons

Types of ions:
- Cation: loses electrons → positive charge (Q>0).
- Anion: gains electrons → negative charge (Q<0).

## 6. Relationship with the periodic table

- The periodic table is ordered by atomic number (Z).
- Each element has a unique Z.
- Elements in the same group have similar properties.

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## 7. Summary

❗ SUPER SUMMARY

- Atoms are made up of protons (+) and neutrons in the nucleus, and electrons (-) in the electron cloud.
- Z = number of protons.
- A = protons + neutrons.
- Q = protons - electrons. Ions have charge:
  - Cation: atom with Q > 0
  - Anion: atom with Q < 0
- Isotopes differ in the number of neutrons.

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## 8. Exercises

### 1. Complete the following table:

| Element | Z | A | Protons | Neutrons | Electrons |
|---|---|---|---|---|---|
| X | 8 | 16 | ? | ? | ? |
| Y | ? | 23 | 11 | ? | 11 |
| Z | 17 | ? | ? | 18 | ? |

Based on the table above:
1. Calculate the missing values.
2. Indicate whether the atom is neutral or an ion.
3. If it is an ion, indicate its charge.

## 2. Challenge question

An atom has: - 12 protons - 12 neutrons - 10 electrons

Answer:

- What is Z?
- What is A?
- Is it an ion?
- What is its charge?